



Data-Driven Corporate Credit Assessment: Machine Learning Approaches to Credit Rating Prediction

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Motivation

- Critical role of credit ratings
 - ▶ Investment and capital requirements
 - ▶ Reducing information asymmetry
- Traditional statistical models struggles with nonlinear patterns
- Models balancing accuracy & interpretability



Limitations of Current Models

- Logistic regression
 - ▶ Imbalanced datasets

- Neural networks and ensemble methods
 - ▶ Lack of transparency and interpretability

- Rising demand for explainability



Hybrid Solution

- Structured predictive framework
 - ▶ Hybrid ML design
 - ▶ Integrates high-quality financial data
- Clustering to identify latent data structures
- Segmentation
 - ▶ Noise
 - ▶ Inconsistence



Data Overview & Data Preprocessing

- Taiwan Economic Journal (TEJ)
- Start with 21,767 observations
- Reduction
 - ▶ Remove mid-year (10,753)
 - ▶ Remove samples without TCRI (4,390)
 - ▶ Drop rare rating category C (2)
- Final dataset: 6,622 observations



Data Preprocessing

▣ Missing Values

- ▶ Most affected: credit impairment (36.8%), retention ratio (22.0%), operating leverage (14.4%)
- ▶ Imputed using median
- ▶ Most are not selected by backward selection

Variable	Missing Values	Missing (%)
Expected credit impairment (loss) profit – operating expense	2,437	36.80
Retention ratio	1,455	21.97
Operating leverage	956	14.43
Financial leverage	845	12.76
R&D expense	232	3.50
Interest coverage ratio	191	2.88
Cash flow adequacy ratio	107	1.62
Cash reinvestment ratio	36	0.54
Contingent liabilities to net worth	2	0.03
ROE – comprehensive income	1	0.02
ROA – comprehensive income	1	0.02



Data Preprocessing

- Financial ratios have extreme skewness & heavy tails
 - ▶ Interest expense ratio show high kurtosis
 - ▶ Outliers: $|z\ score| > 3$, heterogeneous firm size and leverage
- Apply log transformation
 - ▶ Reduces influence of extreme values
 - ▶ Normalizes variables
- Scaling predictors
 - ▶ Prevents large-magnitude variables from dominating training
 - ▶ Ensures comparability



TCRI Distribution

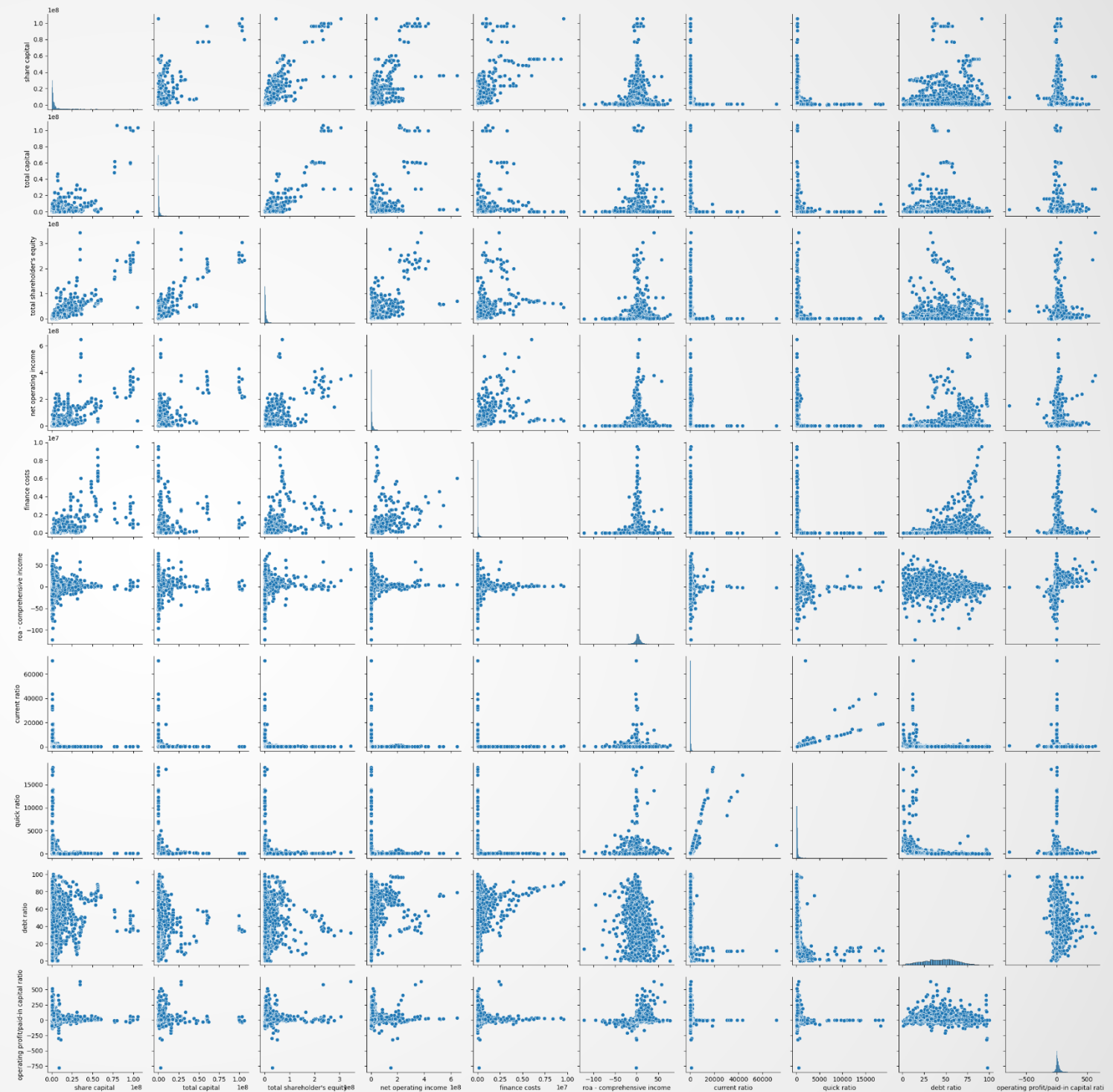
- 5~9 and D: moderate to high risk
- Concentration in ratings 5 and 6
- Imbalance: low prevalence of rating 9 and default status

TCRI Rating	Count	Percentage (%)
5	2054	31.01
6	2161	32.62
7	1389	20.97
8	667	10.07
9	292	4.41
D	59	0.89
C	2	0.03



TCRI Visualization: Pairwise Scatter Plot

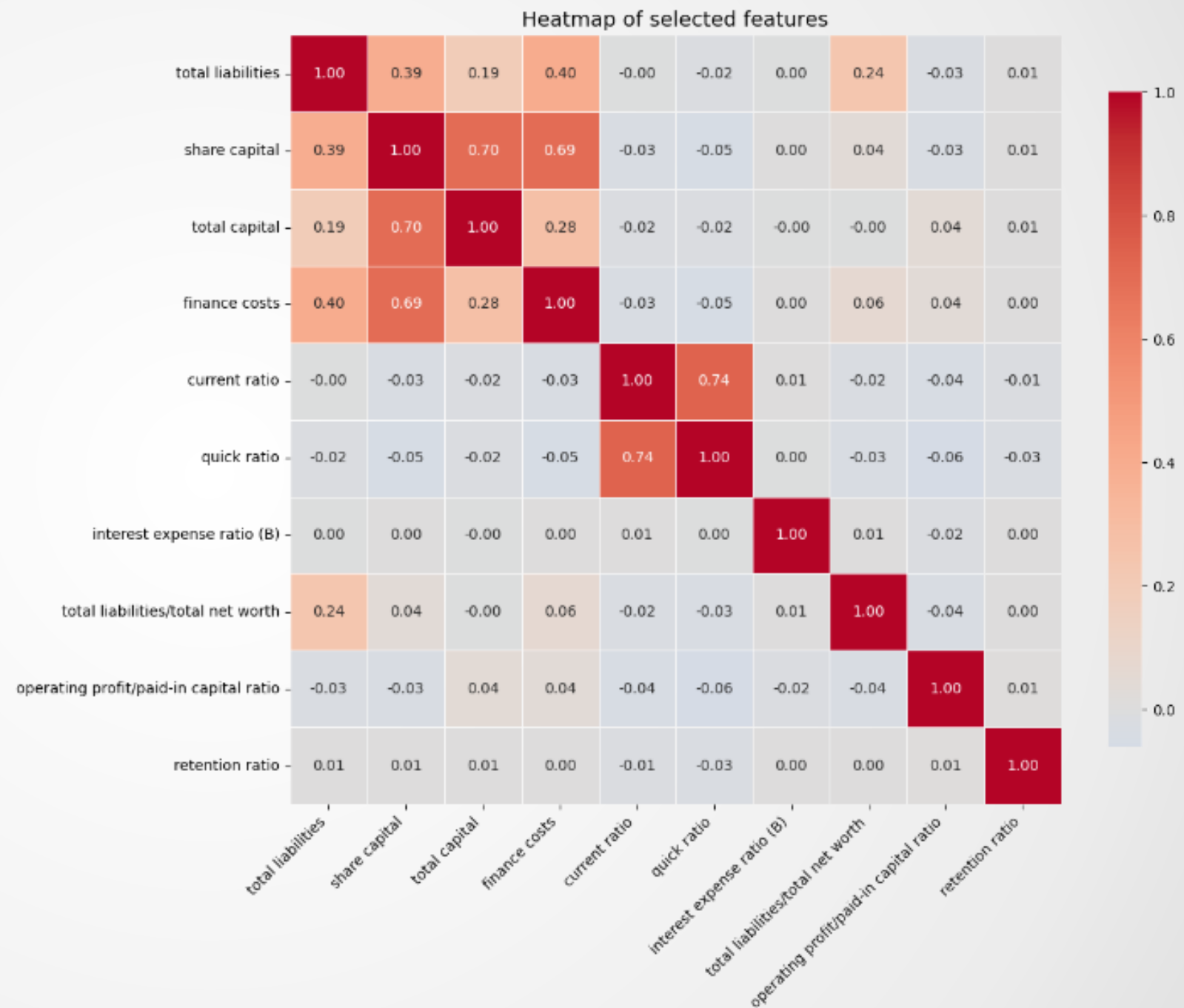
- 10 predictors:
 - ▶ Untransformed form
- Observations:
 - ▶ Visible outliers in interest expense and operating profit
 - ▶ Profitability cluster at higher ratings
 - ▶ Extreme leverage and liabilities: lower-rated or defaulted firms



TCRI Visualization: Heatmap

Heatmap

- ▶ Strong positive correlations among size (share capital, total capital, finance costs)
- ▶ High correlation (0.74) between current and quick ratios
- ▶ Other variables show weak correlations



Benchmark Model: Multinomial Logistic Regression

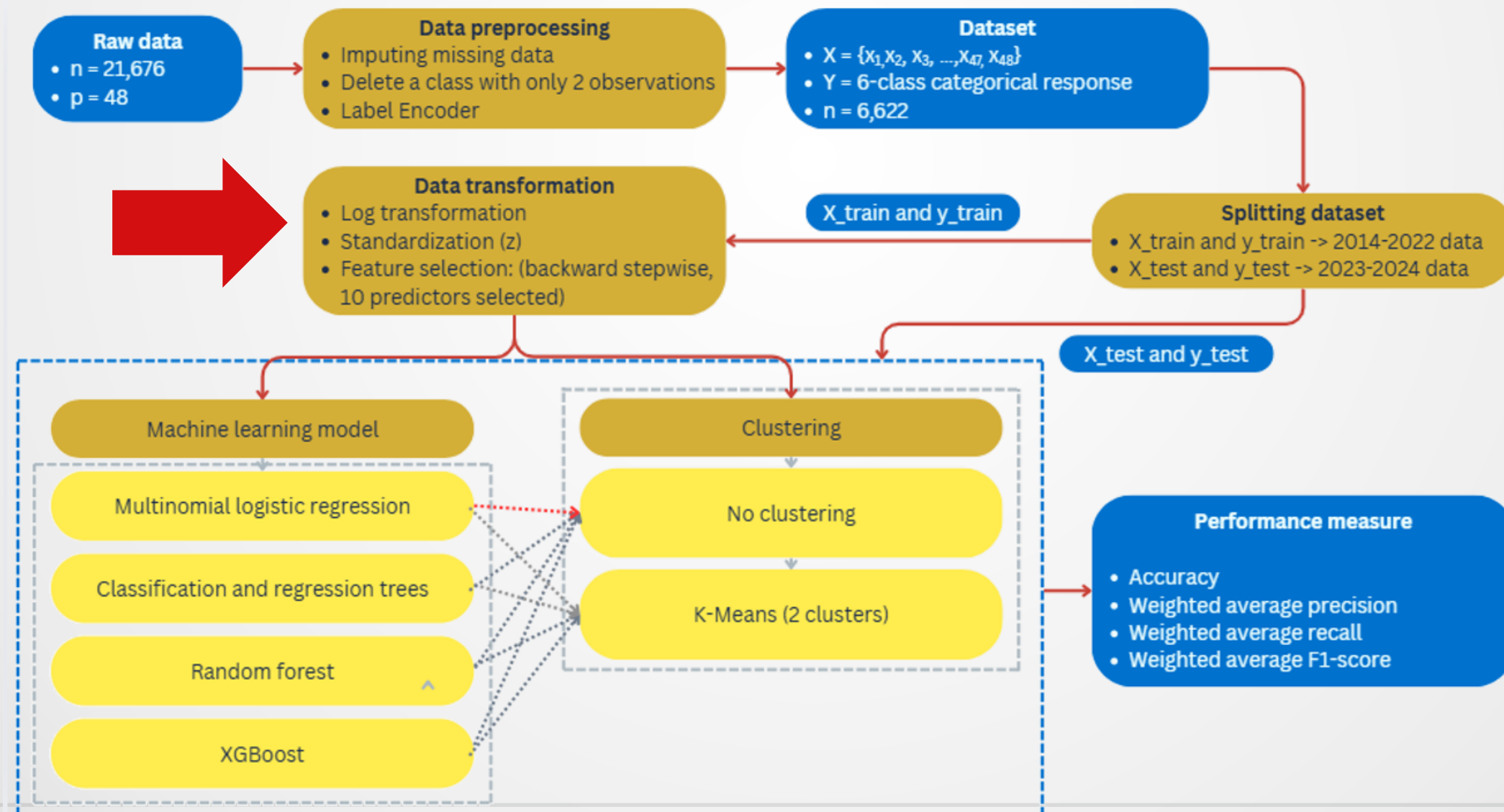
- Standard approach in credit scoring
- Interpretable and regulatory accepted
 - ▶ Accepted in banking and compliance fields
 - ▶ Coefficients -> intuitive odds ratios
- Robust baseline
 - ▶ Transparency for comparison
 - ▶ Moderate data imbalance
 - ▶ Identify monotonic relationships between financial variables and credit risk



Research Design: Data Preprocessing

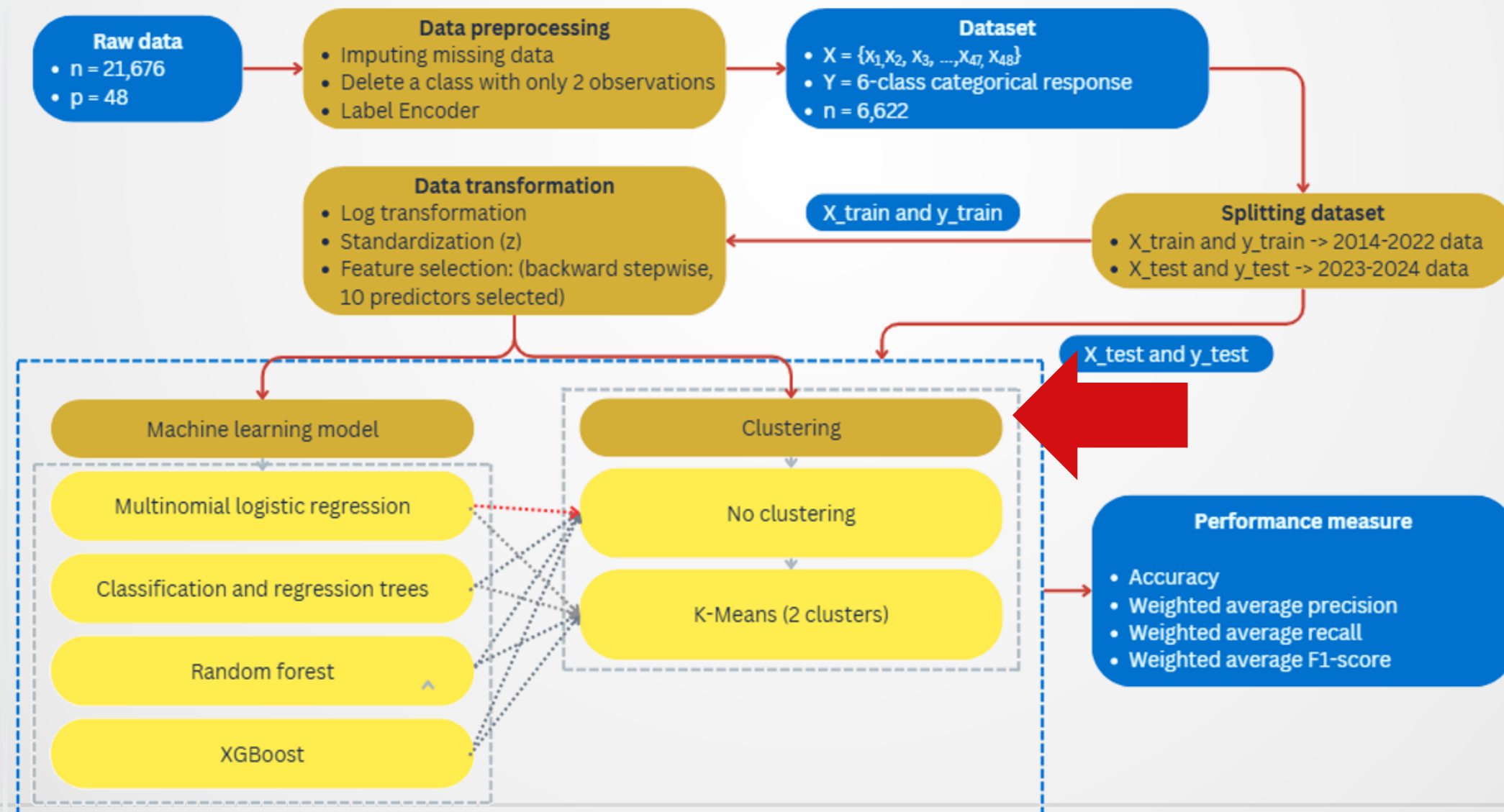
□ Data reduction, log transformation, filling missing value

□ Split dataset



Research Design: Clustering

- Un-supervised clustering: K means
 - ▶ Capture latent structures, reduce noise
- Overcome unrepresentative samples and class imbalance



Feature Selection

- Forward and backward selection were applied
 - ▶ Backward selection provides stronger performance
 - ▶ Other scores follow similar patterns

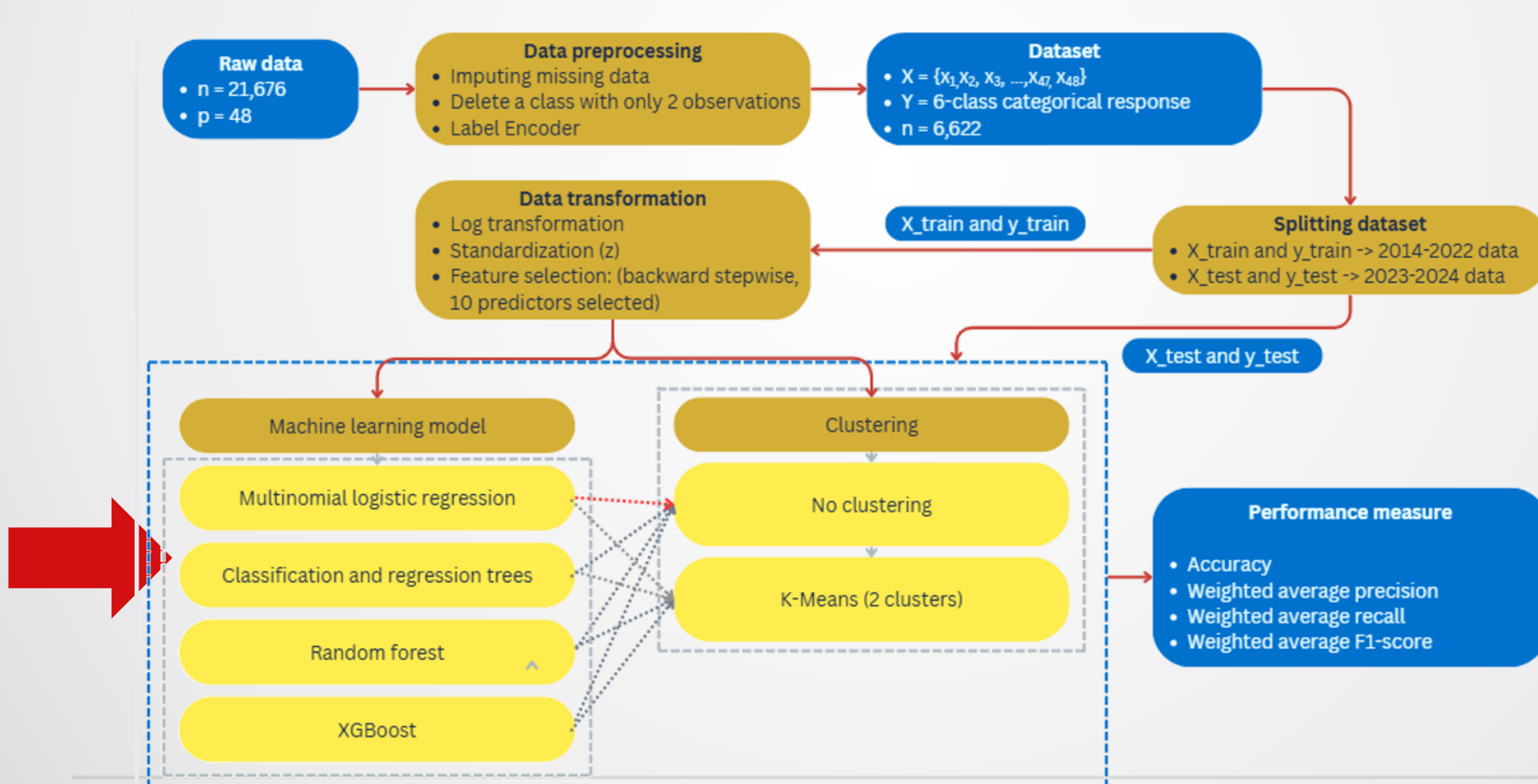
- Reducing predictor set vs. Accuracy

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
MLR: Forward 10 preds	0.51	0.51	0.51	0.50
MLR: Forward 20 preds	0.52	0.51	0.52	0.51
MLR: Forward auto	0.52	0.51	0.52	0.51
MLR: Backward 10 preds	0.55	0.54	0.55	0.54
MLR: Backward 20 preds	0.54	0.53	0.54	0.53
MLR: Backward auto	0.54	0.54	0.54	0.53



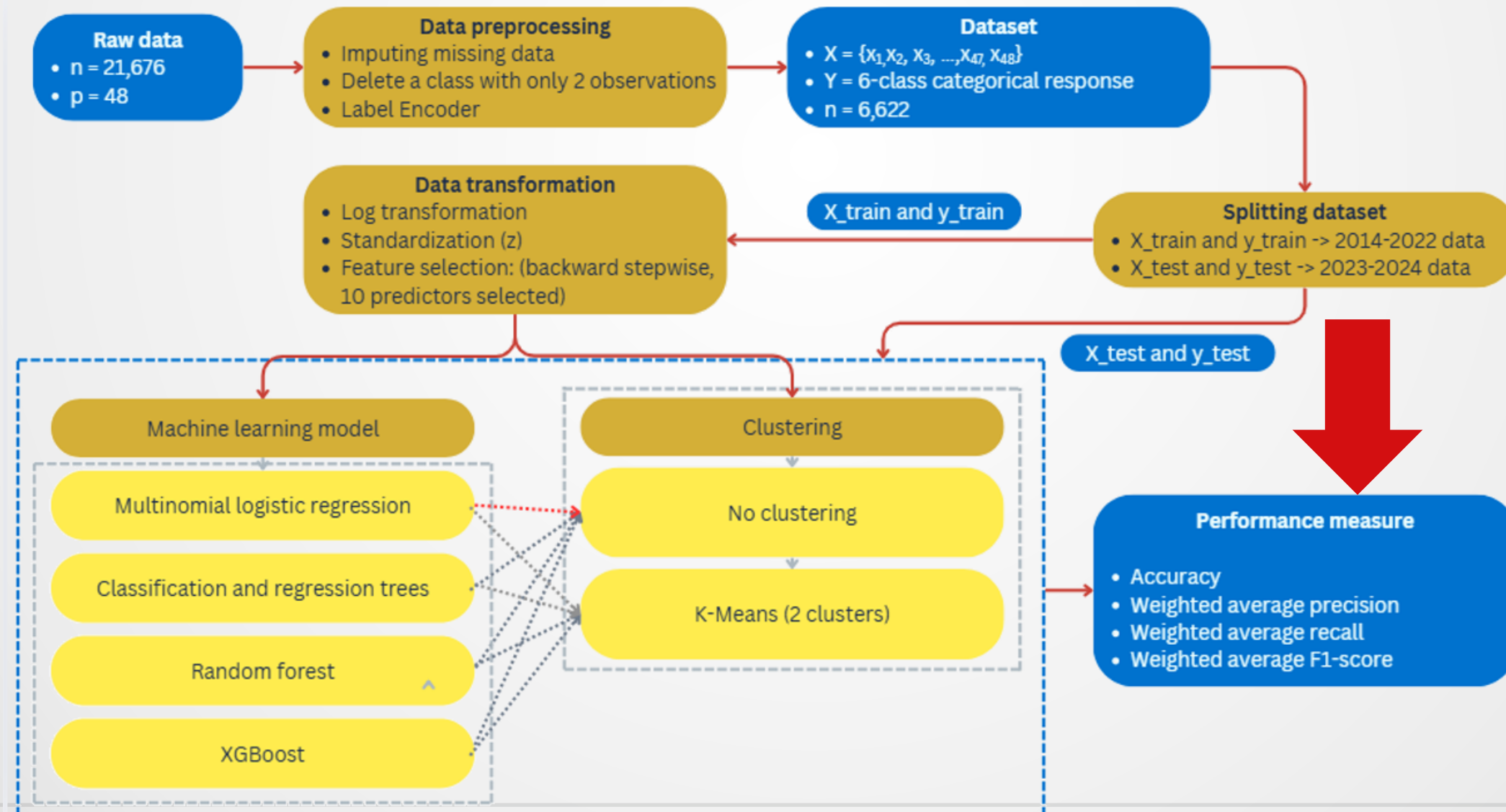
Research Design: Classification within Clusters

- Supervised classifiers:
 - MLR, CART, Random Forest, XGBoost
- Evaluated using cross-validation and test-set accuracy



Research Design: Comparative Benchmarking

- Each cluster-based model against its counterpart
- Performance measures
 - Accuracy, weighted precision, weighted recall, and weighted F1- score



Benchmark Performance

- ▣ Standalone MLR: 0.54 accuracy
- ▣ MLR behavior across clusters: inconsistent predictive gain
 - ▶ Improves slightly in cluster 1
 - ▶ Cluster 1 contains structurally consistent firms

Group	Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
No cluster	Multinomial Logistic Regression	0.54	0.54	0.54	0.53
No cluster	CART	0.48	0.48	0.48	0.48
No cluster	RandomForest	0.62	0.62	0.62	0.62
No cluster	XGBoost	0.63	0.63	0.63	0.63
Cluster 1	Multinomial Logistic Regression	0.58	0.57	0.58	0.57
Cluster 1	CART	0.49	0.50	0.49	0.49
Cluster 1	RandomForest	0.60	0.60	0.60	0.60
Cluster 1	XGBoost	0.63	0.63	0.63	0.63
Cluster 2	Multinomial Logistic Regression	0.53	0.52	0.53	0.51
Cluster 2	CART	0.49	0.50	0.49	0.49
Cluster 2	RandomForest	0.61	0.62	0.61	0.61
Cluster 2	XGBoost	0.62	0.62	0.62	0.62



Model Performance

- CART underperforms: weak nonlinear modeling
- Random Forest is stable across clusters
- XGBoost consistently outperforms (≈ 0.63) without clustering
- Ensemble methods capture latent structure internally

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Cluster 2	CART	0.49	0.50	0.49	0.49
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Role of Clustering

- Clustering does not enhance predictive performance for TCRI
 - ▶ MLR performance improves (0.54 -> 0.58) within cluster 1
 - ▶ Gains are inconsistent across clusters and negligible for ensemble methods
- Clustering may serve subgroup interpretive purposes

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Cluster 2	CART	0.49	0.50	0.49	0.49
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Empirical Findings

- Superiority of ensemble methods
 - ▶ XGBoost: highest overall accuracy
 - ▶ Captures nonlinear financial interactions
- Clustering provides negligible improvement
 - ▶ MLR improves only in Cluster
 - ▶ Random Forest and XGBoost: stable with or without clustering
- Feature selection enhances interpretability
 - ▶ Backward selection removes redundant predictors
 - ▶ Retained variables reflect key financial factors



Conclusions

- Structured framework for credit rating prediction
 - ▶ Data preprocessing, feature selection, and multiple classifiers
 - ▶ Ensemble methods achieve strongest overall performance
- Balance interpretability and predictive power
 - ▶ MLR: transparent, has limited flexibility
 - ▶ Nonlinear methods: accurate, depend on data characteristics
- Future work
 - ▶ Alternative classifiers: LightGBM, CatBoost, or neural network
 - ▶ Incorporate non-financial variables: ESG indicators, macroeconomic factors, or textual data





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